

# Machine Learning Algorithms Laboratory

## Experiment 3 Report

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| Experiment No.   | 3                                                                            |
| Experiment Title | Regression Analysis using Linear and Regularized Models<br><a href="#">1</a> |
| Student Name     | Gautham S                                                                    |
| Register Number  | 3122247001016                                                                |
| Faculty          | Dr. Poreddy Ajay Kumar Reddy                                                 |
| Submission Date  |                                                                              |

## 1 Aim and Objective

The aim of this experiment is to implement Linear Regression and its regularized versions (Ridge, Lasso and Elastic Net) to predict the loan amount that will be sanctioned to a customer. The models are compared using different regression metrics, and overfitting/underfitting behaviour along with the bias-variance trade-off is analyzed.

## 2 Dataset Description

| Dataset            | Description                                                                               |
|--------------------|-------------------------------------------------------------------------------------------|
| Dataset Name       | Loan Amount Prediction Dataset                                                            |
| Dataset Source     | Kaggle - Predict Loan Amount Data                                                         |
| Number of Samples  | 30000 (29322 used after removing rows with missing target)                                |
| Number of Features | 23 raw features (numerical + categorical), 45 after one-hot encoding                      |
| Target Variable    | Loan Sanction Amount (USD)                                                                |
| Missing Values     | Present in several columns (Type of Employment, Property Age, Income, Credit Score, etc.) |
| Train-Test Split   | 80% Train (23457 samples), 20% Test (5865 samples)                                        |

## 3 Preprocessing Steps

The dataset had a lot of missing values and also some invalid entries. The following steps were performed:

- Some columns like Current Loan Expenses, Co-Applicant, Property Price and the target column itself had the value -999 in place of missing data, so these were first replaced with NaN.

- Rows where the target (Loan Sanction Amount) was missing were dropped since they cannot be used for training.
- Customer ID, Name and Property ID columns were dropped as they are just identifiers and not useful features.
- Missing values in numerical columns were filled with the median, and missing values in categorical columns were filled with the mode.
- Categorical columns were converted into numerical form using one-hot encoding.
- Numerical features were standardized using `StandardScaler` before training, since Ridge, Lasso and Elastic Net are sensitive to feature scale.

```
cols_with_invalid = ["Current Loan Expenses (USD)", "Co-Applicant",
                    "Property Price", "Loan Sanction Amount (USD)"]

df[cols_with_invalid] = df[cols_with_invalid].replace(-999, np.nan)
df.dropna(subset=["Loan Sanction Amount (USD)"], inplace=True)
df.drop(columns=["Customer ID", "Name", "Property ID"], inplace=True)

X[num_cols] = X[num_cols].fillna(X[num_cols].median())
X[cat_cols] = X[cat_cols].fillna(X[cat_cols].mode().iloc[0])

X = pd.get_dummies(X, columns=cat_cols, drop_first=True)
```

## 4 Implementation Details

### 4.1 Models Used

- **Linear Regression** - baseline model with no regularization.
- **Ridge Regression** - adds an L2 penalty to the loss function, which shrinks the coefficients but does not make them exactly zero.
- **Lasso Regression** - adds an L1 penalty, which can shrink some coefficients exactly to zero, effectively performing feature selection.
- **Elastic Net** - combines both L1 and L2 penalty using a mixing ratio (`l1_ratio`).

### 4.2 Experimental Setup

---

|                      |                  |
|----------------------|------------------|
| Python Version       | 3.11             |
| Scikit-Learn Version | 1.4              |
| IDE                  | Jupyter Notebook |
| Operating System     | Windows 11       |

---

### 4.3 Hyperparameter Tuning

Ridge, Lasso and Elastic Net all have a regularization strength parameter  $\alpha$ , and Elastic Net additionally has an `l1_ratio` parameter. These were tuned using `GridSearchCV` with 5-fold cross validation, optimizing for R2 score.

```
ridge_params = {"alpha": [0.01, 0.1, 1, 10, 100]}

ridge_grid = GridSearchCV(Ridge(), ridge_params, cv=5, scoring="r2")
ridge_grid.fit(X_train_scaled, y_train)

print(ridge_grid.best_params_)
print(ridge_grid.best_score_)
```

Lasso used  $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$  and Elastic Net used  $\alpha \in \{0.01, 0.1, 1, 10\}$  with  $l1\_ratio \in \{0.2, 0.5, 0.8\}$ , in the same manner.

## 5 Visualizations

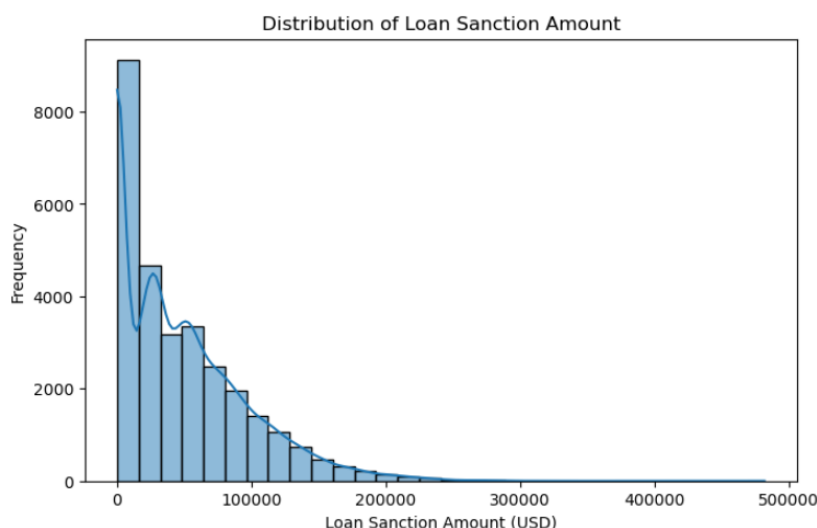


Figure 1: Distribution of Loan Sanction Amount

**Inference:** The target variable is right skewed, with most loan amounts concentrated on the lower end and a long tail of higher values. This kind of skew can make it a bit harder for a linear model to fit well across the full range of the target.

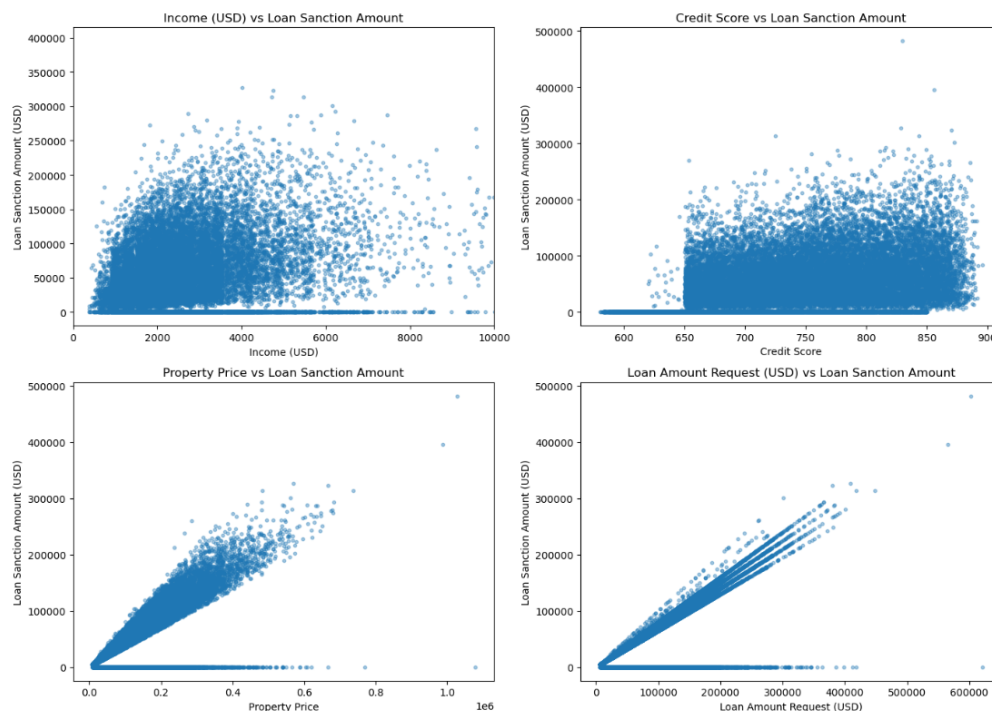


Figure 2: Feature vs Target Scatter Plots

**Inference:** Loan Amount Request shows a clear positive linear relationship with the Loan Sanction Amount, which is expected since the sanctioned amount usually depends directly on the requested amount. Income and Property Price also show a mild positive trend, while Credit Score does not show a very strong pattern.

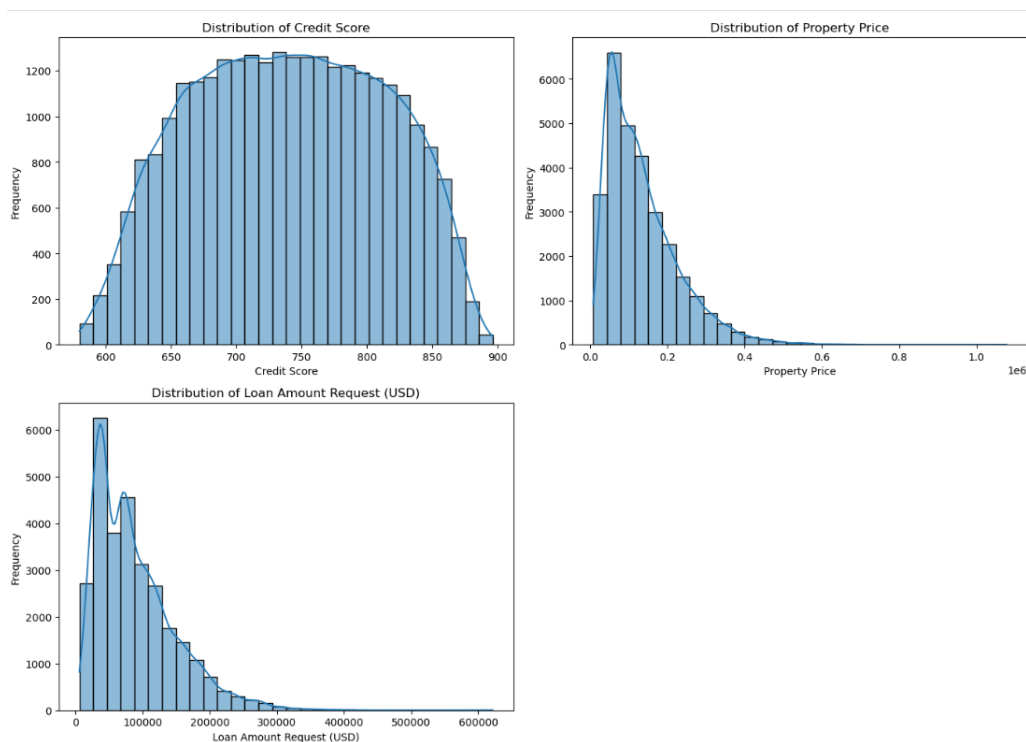


Figure 3: Distribution of Important Numerical Features

**Inference:** Credit Score is roughly spread across its range, while Property Price and Loan Amount Request are both right skewed similar to the target variable, with a few very high value outliers.

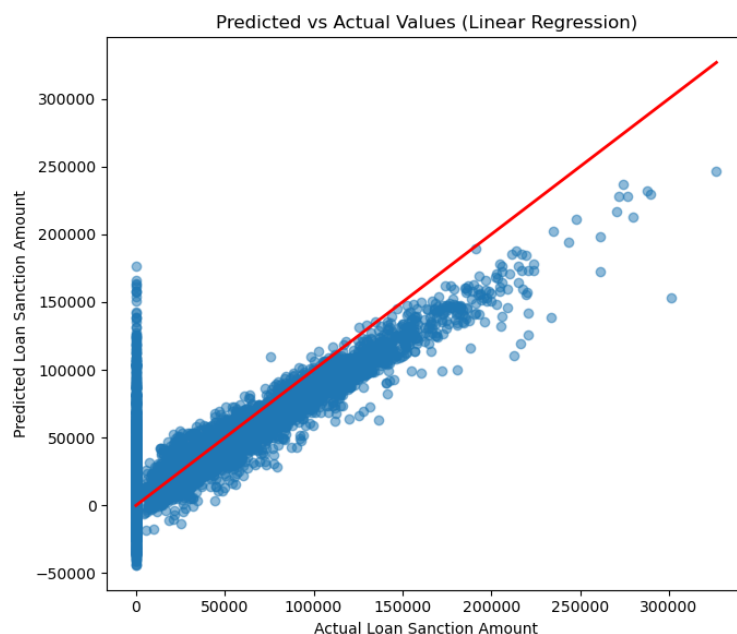


Figure 4: Predicted vs Actual Values (Linear Regression)

**Inference:** Most of the points lie reasonably close to the red diagonal line which represents a perfect prediction, but there is noticeable spread, especially for higher loan amounts. This means the model captures the overall trend but is less accurate for larger loan values.

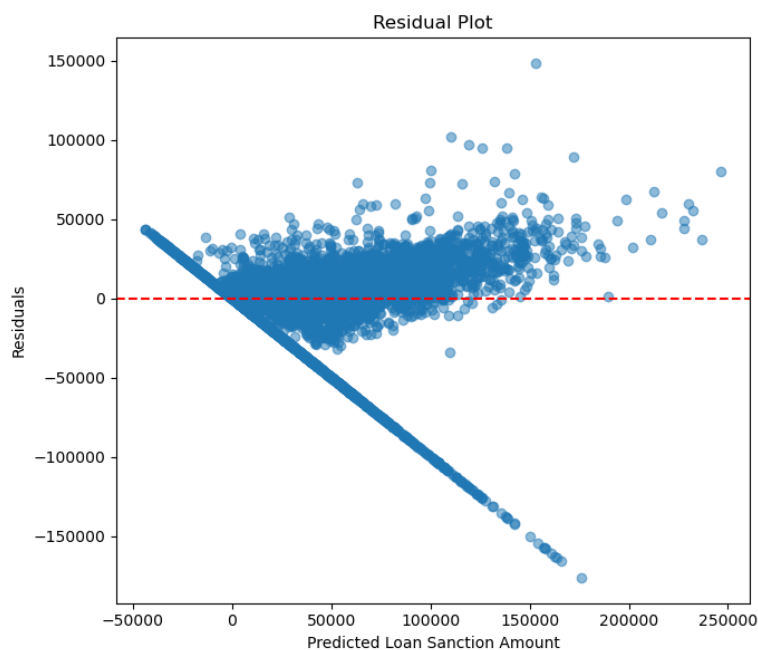


Figure 5: Residual Plot (Linear Regression)

**Inference:** The residuals are mostly centered around zero but the spread increases

for higher predicted values, forming a slight funnel shape. This indicates a bit of heteroscedasticity, meaning the model's error is not perfectly constant across all prediction ranges.

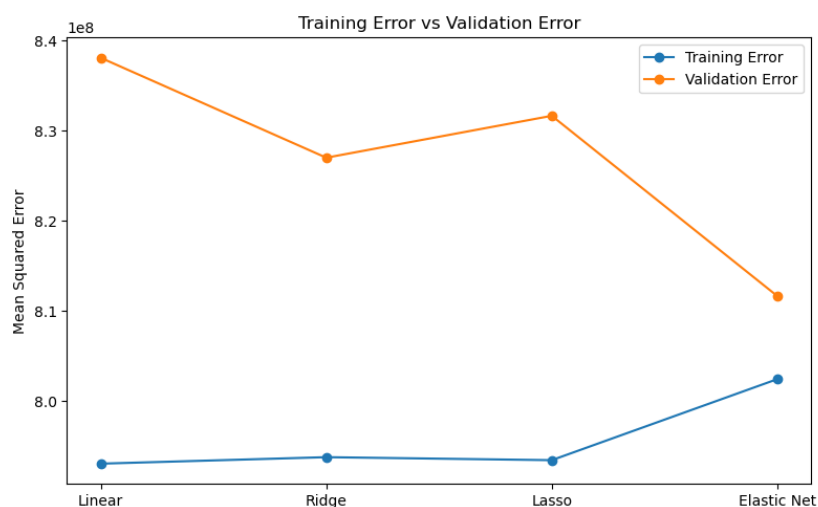


Figure 6: Training Error vs Validation Error

**Inference:** The training and validation errors are quite close to each other for all four models, which suggests the models are not overfitting much on this dataset. Elastic Net has a slightly lower validation error compared to the others.

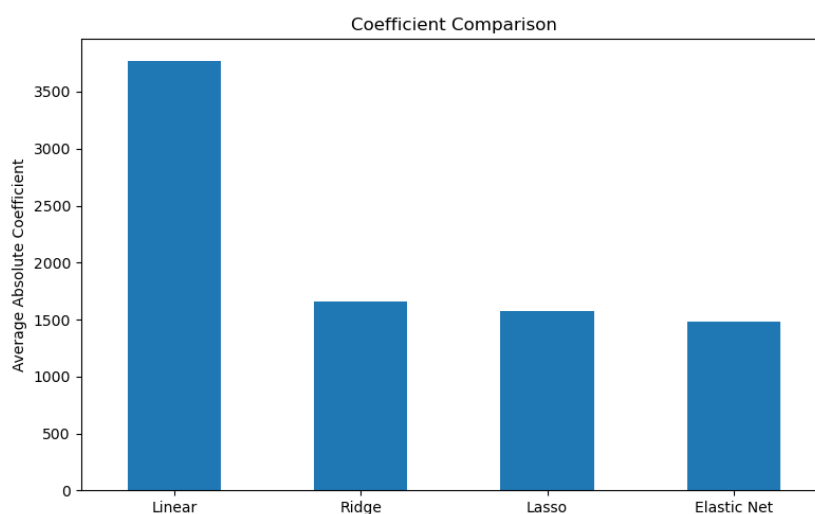


Figure 7: Coefficient Comparison Across Models

**Inference:** The average absolute coefficient value drops noticeably from Linear Regression to Ridge, Lasso and Elastic Net, showing that regularization is shrinking the coefficients as expected. Lasso shrinks some coefficients to exactly zero (for example Income (USD) becomes almost 0), effectively removing those features.

## 6 Performance Tables

### 6.1 Hyperparameter Tuning Summary

| Model                  | Search Method | Best Parameters             | Best CV R2 |
|------------------------|---------------|-----------------------------|------------|
| Ridge Regression       | Grid Search   | alpha = 100                 | 0.6471     |
| Lasso Regression       | Grid Search   | alpha = 10                  | 0.6452     |
| Elastic Net Regression | Grid Search   | alpha = 0.1, l1_ratio = 0.5 | 0.6534     |

Table 2: Hyperparameter Tuning Summary

### 6.2 Cross-Validation Performance (K=5)

| Model                  | MAE      | MSE    | RMSE     | R2     |
|------------------------|----------|--------|----------|--------|
| Linear Regression      | 18903.35 | 8.38e8 | 28949.45 | 0.6425 |
| Ridge Regression       | 18915.35 | 8.27e8 | 28757.62 | 0.6471 |
| Lasso Regression       | 18895.34 | 8.32e8 | 28838.26 | 0.6452 |
| Elastic Net Regression | 19077.20 | 8.12e8 | 28489.76 | 0.6534 |

Table 3: Cross-Validation Performance

### 6.3 Test Set Performance

| Model                  | MAE      | MSE    | RMSE     | R2     | Training Time (s) |
|------------------------|----------|--------|----------|--------|-------------------|
| Linear Regression      | 18761.84 | 7.70e8 | 27756.40 | 0.6575 | 0.0133            |
| Ridge Regression       | 18783.05 | 7.71e8 | 27770.98 | 0.6571 | 0.2050            |
| Lasso Regression       | 18755.72 | 7.70e8 | 27753.43 | 0.6576 | 56.3093           |
| Elastic Net Regression | 19009.39 | 7.83e8 | 27976.20 | 0.6520 | 9.0812            |

Table 4: Test Set Performance

**Inference:** All four models perform quite similarly on the test set, with R2 values between 0.65 and 0.66. Lasso and Linear Regression gave marginally the best R2, but Lasso took by far the longest time to train (around 56 seconds) since it uses coordinate descent which needed more iterations to converge here, while Linear Regression trains almost instantly using a closed form solution.

### 6.4 Effect of Regularization on Coefficients

| Feature                   | Linear    | Ridge    | Lasso    | Elastic Net |
|---------------------------|-----------|----------|----------|-------------|
| Property Age              | -46437.28 | -940.61  | -17.97   | -99.29      |
| Income (USD)              | 46415.53  | 912.95   | -0.00    | 61.36       |
| Loan Amount Request (USD) | 33139.27  | 31419.58 | 33113.11 | 22863.95    |

Table 5: Coefficient Comparison (Top 3 Features)

**Inference:** Property Age and Income (USD) have huge coefficients in plain Linear Regression, which is a bit unstable and probably means they are correlated with other features. Regularization brings these values down drastically. Lasso even reduces Income (USD)'s coefficient to almost exactly zero, meaning it decided this feature was not important once other correlated features were considered. Loan Amount Request stays fairly large and important across all models, which makes sense since it is the most directly related feature to the target.

## 7 Overfitting and Underfitting Analysis

From the training vs validation error plot, the training and validation errors are close to each other for all four models, so there isn't strong evidence of overfitting here. This could be because the dataset has a decent number of samples (about 23000 for training) compared to the number of features (45), so a linear model does not have too much room to overfit.

As the regularization strength ( $\alpha$ ) increases, the model coefficients shrink more, which reduces variance but can increase bias if  $\alpha$  is too high. In this experiment, the best  $\alpha$  values found by GridSearchCV (Ridge: 100, Lasso: 10) were on the higher end of the search range, which suggests that at least some regularization does help slightly, but the  $R^2$  scores did not change by a large amount between the plain and regularized models. This means the baseline Linear Regression was not overfitting a lot to begin with, so regularization only gave a small improvement in generalization.

## 8 Bias-Variance Analysis

Linear Regression, being unregularized, can have higher variance since it will try to fit every feature's coefficient without any constraint. This was seen in the large and possibly unstable coefficient values for Property Age and Income compared to the other models. Ridge Regression reduces this variance by shrinking all coefficients smoothly, without setting any of them exactly to zero, which is why it is generally more stable when features are correlated with each other.

Lasso, on the other hand, performs feature selection by pushing some coefficients exactly to zero (for example Income (USD) became almost 0), which increases the model's bias slightly on those features but can be useful when a lot of features are irrelevant or redundant. Elastic Net tries to balance both behaviours since it uses a mix of L1 and L2 penalty, and in this experiment it gave the best cross-validation  $R^2$  (0.6534) among the regularized models, suggesting a reasonable trade-off between bias and variance for this dataset.

## 9 Observations and Conclusion

All four regression models gave fairly similar performance on this dataset, with  $R^2$  scores between about 0.64 and 0.66 for both cross-validation and test data. This means the linear relationship captured by these models explains around 65% of the variance in the loan sanction amount, and adding regularization did not change performance drastically since the baseline model was not overfitting significantly in the first place.

Lasso Regression gave a marginally better test  $R^2$  than the other models, and also has the added benefit of automatic feature selection (as seen in the Income (USD) coefficient going to zero). However, its training time was significantly higher than Linear Regression, Ridge and Elastic Net. Elastic Net gave the best cross-validation  $R^2$ , indicating it might generalize best to unseen data among the regularized models.

Overall, if training time matters more than a very small performance gain, Ridge or plain Linear Regression can be preferred here. If feature selection and slightly better generalization is important, Lasso or Elastic Net would be a better choice. This experiment gave a good hands-on understanding of how regularization affects model coefficients, training time, and the bias-variance trade-off.

## 10 References

1. Scikit-learn Documentation: Linear Models. [https://scikit-learn.org/stable/modules/linear\\_model.html](https://scikit-learn.org/stable/modules/linear_model.html)
2. Scikit-learn Documentation: Hyperparameter Optimization. [https://scikit-learn.org/stable/modules/grid\\_search.html](https://scikit-learn.org/stable/modules/grid_search.html)
3. Loan Amount Prediction Dataset, Kaggle.